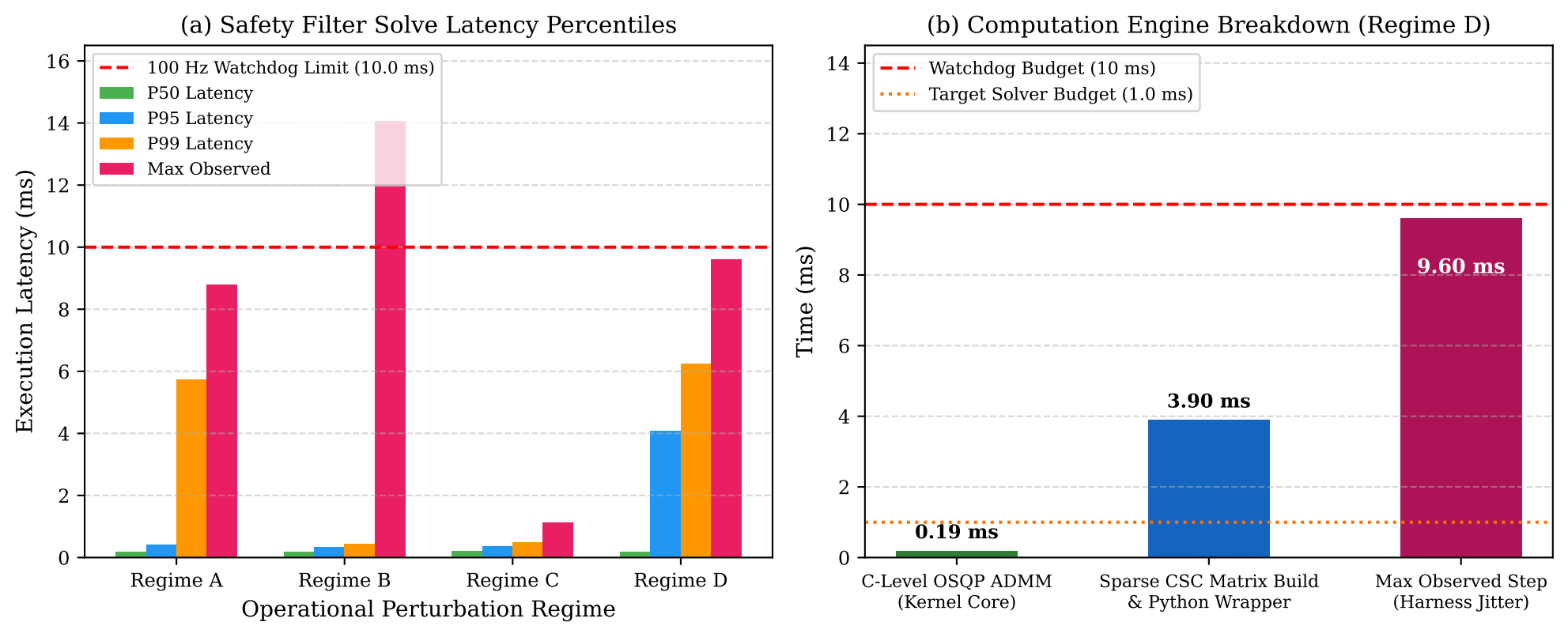
(a) Safety Filter Solve Latency Percentiles
100 Hz Watchdog Limit (10.0 ms)
P50 Latency
P95 Latency
P99 Latency
Max Observed
Execution Latency (ms)
0
2
4
6
8
10
12
14
16
Regime A
Regime B
Regime C
Regime D
Operational Perturbation Regime
(b) Computation Engine Breakdown (Regime D)
Watchdog Budget (10 ms)
Target Solver Budget (1.0 ms)
Time (ms)
0
2
4
6
8
10
12
14
0.19 ms
3.90 ms
9.60 ms
C-Level OSQP ADMM
(Kernel Core)
Sparse CSC Matrix Build
& Python Wrapper
Max Observed Step
(Harness Jitter)